## Types of acid & base

| | Acid | Base |
|---|---|---|
| Arrhenius | | |
| Bronsted | | |
| Lewis | | |

[For Bronsted acids/ base]
Identify the weak acid/ base & its conjugate acid /base

NaOH + HCN → NaCN + $H_2O$

*Note:*

- conjugate acid-base pairs always differ in formula by $H^+$

## $K_a$ & $K_b$ expressions

Write the equation & $K_a$ expression for the weak acid: HF

Write the equation & $K_b$ expression for its conjugate base

Relationship between $K_a$ of acid and $K_b$ of its conjugate base

## Equations

| | | |
|---|---|---|
| *p*H | = | |
| *p*OH | = | |
| $pK_a$ | = | $-\lg K_a$ |
| $pK_b$ | = | $-\lg K_b$ |
| $pK_w$ | = | $-\lg K_w$ |

| | | |
|---|---|---|
| $K_w$ | = | |
| $pK_w$ | = | |

*Note:*

- *$K_w$ is affected by temperature*
- *$pK_w$ = 14 only at ______ °C*

## Identifying types of solutions

Types of solutions

- strong acid/ strong base
- weak acid/ weak base
- acidic salt/ basic salt/ neutral salt
- buffer

➢ Deduce the type of solution formed after mixing

| | |
|---|---|
| 0.1 mol of NaOH added to a solution containing 0.1 mol HF<br>$OH^- + HF \rightarrow H_2O + F^-$ | 0.1 mol of NaOH added to a solution containing 1.0 mol HF |
| 0.5 mol of NaOH added to a solution containing 1.0 mol HF and 0.5 mol $F^-$ | 0.5 mol of $HNO_3$ added to a solution containing 1.0 mol HF and 0.5 mol $F^-$ |

# pH of solutions

## 1 Strong acid / base

Calculate the pH of 0.1 mol $dm^{-3}$ $HNO_3$(aq) solution

pH = 1.00

Calculate the pH of 0.02 mol $dm^{-3}$ NaOH(aq) solution

pH = 12.3

## 2 Weak acid / base

Calculate the *p*H of 0.1 mol $dm^{-3}$ $CH_3COOH$(aq) solution
*($pK_a$ of $CH_3COOH$ = 4.8)*

$[H^+] = \sqrt{K_a\,[\text{weak acid}]}$

$[H^+] = \sqrt{K_a\,[CH_3COOH]}$

$= \sqrt{(10^{-4.8})[0.1]}$

$= 1.26 \times 10^{-3}$ mol $dm^{-3}$

pH $= -\lg[H^+]$

$= -\lg[1.26 \times 10^{-3}]$

$= 2.90$

Calculate the pH of 0.2 mol $dm^{-3}$ $NH_3$(aq) solution
*($pK_b$ of $NH_3$ = 4.7)*

pH = 11.3

## 3 Salt

Calculate the pH of 0.5 mol $dm^{-3}$ $CH_3COONa$(aq) solution
*($pK_a$ of $CH_3COOH$ = 4.8 ⇒ $pK_b$ of $CH_3COO^-$ = ______ )*

pH = 9.25

Calculate the pH of 1.00 mol $dm^{-3}$ $NH_4Cl$(aq) solution
*($pK_b$ of $NH_3$ = 4.7 ⇒ $pK_a$ of $NH_4^+$ = ______ )*

pH = 4.65

## 4 Buffer

Calculate the pH of a solution containing 0.1 mol $dm^{-3}$ $CH_3COOH$ & 0.5 mol $dm^{-3}$ $CH_3COONa$. *($pK_a$ of $CH_3COOH$ = 4.8)*

pH = 5.50

Calculate the pH of a 0.5 $dm^3$ solution containing 0.1 mol $NH_3$ & 0.5 mol $NH_4Cl$. *($pK_b$ of $NH_3$ = 4.7)*

pH = 8.60

## Types of acid & base

| | Acid | Base |
|---|---|---|
| Arrhenius | release $H^+$ in (aq) | release $OH^-$ in (aq) |
| Bronsted | proton donor | proton acceptor |
| Lewis | electron pair acceptor | electron pair donor |

[For Bronsted acids/ base]
Identify the weak acid/ base & its conjugate acid /base

$NaOH + HCN \rightarrow NaCN + H_2O$

HCN: weak acid; NaCN: conjugate base

*Note:*

- conjugate acid-base pairs always differ in formula by $H^+$

## $K_a$ & $K_b$ expressions

Write the equation & $K_a$ expression for the weak acid: HF

$HF \rightleftharpoons H^+ + F^-$  $K_a = \frac{[H^+][F^-]}{[HF]}$

Write the equation & $K_b$ expression for its conjugate base

$F^- + H_2O \rightleftharpoons HF + OH^-$  $K_b = \frac{[HF][OH^-]}{[F^-]}$

Relationship between $K_a$ of acid and $K_b$ of its conjugate base

$K_a \times K_b = K_w$

## Equations

| | | |
|---|---|---|
| $pH$ | = | $-\lg [H^+]$ |
| $pOH$ | = | $-\lg [OH^-]$ |
| $pK_a$ | = | $-\lg K_a$ |
| $pK_b$ | = | $-\lg K_b$ |
| $pK_w$ | = | $-\lg K_w$ |

| | | |
|---|---|---|
| $K_w$ | = | $[H^+][OH^-]$ |
| $pK_w$ | = | $pH + pOH$ |

*Note:*

- *$K_w$ is affected by temperature*
- $pK_w$ = 14 only at 25 °C

## Identifying types of solutions

Types of solutions

- strong acid/ strong base
- weak acid/ weak base
- acidic salt/ basic salt/ neutral salt
- buffer

➢ Deduce the type of solution formed after mixing

| | |
|---|---|
| 0.1 mol of NaOH added to a solution containing 0.1 mol HF<br>$OH^- + HF \rightarrow H_2O + F^-$<br>resultant solⁿ: 0.1 mol NaF ⇒ basic salt | 0.1 mol of NaOH added to a solution containing 1.0 mol HF<br>$OH^- + HF \rightarrow H_2O + F^-$<br>resultant solⁿ: 0.9 mol HF, 0.1 mol $F^-$ ⇒ buffer |
| 0.5 mol of NaOH added to a solution containing 1.0 mol HF and 0.5 mol $F^-$<br>$OH^- + HF \rightarrow H_2O + F^-$<br>resultant solⁿ: 0.5 mol HF, 1.0 mol $F^-$ ⇒ buffer | 0.5 mol of $HNO_3$ added to a solution containing 1.0 mol HF and 0.5 mol $F^-$<br>$H^+ + F^- \rightarrow HF$<br>resultant solⁿ: 1.5 mol HF ⇒ weak acid |

# pH of solutions

## 1 Strong acid / base

Calculate the pH of 0.1 mol dm$^{-3}$ $HNO_3$(aq) solution

$HNO_3 \rightarrow H^+ + NO_3^-$

$[H^+] = 0.1$ mol dm$^{-3}$

$pH = -\lg [H^+]$
$= -\lg [0.1]$
$= 1.00$

pH = 1.00

Calculate the pH of 0.02 mol dm$^{-3}$ NaOH(aq) solution

$NaOH \rightarrow Na^+ + OH^-$

$[OH^-] = 0.02$ mol dm$^{-3}$

$pOH = -\lg [OH^-]$
$= -\lg [0.02]$
$= 1.70$

$pH = 14 - 1.70$
$= 12.3$

pH = 12.3

## 2 Weak acid / base

Calculate the pH of 0.1 mol dm$^{-3}$ $CH_3COOH$(aq) solution
*($pK_a$ of $CH_3COOH$ = 4.8)*

$[H^+] = \sqrt{K_a [\text{weak acid}]}$

$[H^+] = \sqrt{K_a [CH_3COOH]}$
$= \sqrt{(10^{-4.8})[0.1]}$
$= 1.26 \times 10^{-3}$ mol dm$^{-3}$

$pH = -\lg[H^+]$
$= -\lg[1.26 \times 10^{-3}]$
$= 2.90$

Calculate the pH of 0.2 mol dm$^{-3}$ $NH_3$(aq) solution
*($pK_b$ of $NH_3$ = 4.7)*

$[OH^-] = \sqrt{K_b [\text{weak base}]}$

$[OH^-] = \sqrt{K_b [NH_3]}$
$= \sqrt{(10^{-4.7})[0.2]}$
$= 2.00 \times 10^{-3}$ mol dm$^{-3}$

$pOH = -\lg [OH^-]$
$= -\lg [2.00 \times 10^{-3}]$
$= 2.70$

$pH = 14 - 2.70$
$= 11.3$

pH = 11.3

## 3 Salt

Calculate the pH of 0.5 mol dm$^{-3}$ $CH_3COONa$(aq) solution
*($pK_a$ of $CH_3COOH$ = 4.8 ⇒ $pK_b$ of $CH_3COO^-$ = 9.2 )*

$[OH^-] = \sqrt{K_b [\text{weak base}]}$

$[OH^-] = \sqrt{K_b [CH_3COO^-]}$
$= \sqrt{(10^{-9.2})[0.5]}$
$= 1.78 \times 10^{-5}$ mol dm$^{-3}$

note:
- $CH_3COO^-$ is a conjugate base of $CH_3COOH$
- $CH_3COO^-$ is a weak base

$pOH = -\lg [OH^-]$
$= -\lg [1.78 \times 10^{-5}]$
$= 4.75$

$pH = 14 - 4.75$
$= 9.25$

pH = 9.25

Calculate the pH of 1.00 mol dm$^{-3}$ $NH_4Cl$(aq) solution
*($pK_b$ of $NH_3$ = 4.7 ⇒ $pK_a$ of $NH_4^+$ = 9.2 )*

$[H^+] = \sqrt{K_a [\text{weak acid}]}$

$[H^+] = \sqrt{K_a [NH_4^+]}$
$= \sqrt{(10^{-9.3})[1.00]}$
$= 2.24 \times 10^{-5}$ mol dm$^{-3}$

note:
- $NH_4^+$ is a conjugate acid of $NH_3$
- $NH_4^+$ is a weak acid

$pH = -\lg[H^+]$
$= -\lg[2.24 \times 10^{-5}]$
$= 4.65$

pH = 4.65

## 4 Buffer

Calculate the pH of a solution containing 0.1 mol dm$^{-3}$ $CH_3COOH$ & 0.5 mol dm$^{-3}$ $CH_3COONa$. *($pK_a$ of $CH_3COOH$ = 4.8)*

$pH = pK_a + \lg \frac{[\text{salt}]}{[\text{acid}]}$
$= pK_a + \lg \frac{[CH_3COO^-]}{[CH_3COOH]}$
$= 4.8 + \lg \frac{[0.5]}{[0.1]}$
$= 5.50$

pH = 5.50

Calculate the pH of a 0.5 dm$^3$ solution containing 0.1 mol $NH_3$ & 0.5 mol $NH_4Cl$. *($pK_b$ of $NH_3$ = 4.7)*

$pOH = pK_b + \lg \frac{[\text{salt}]}{[\text{base}]}$
$= pK_b + \lg \frac{[NH_4^+]}{[NH_3]}$
$= 4.7 + \lg \frac{[0.5 / 0.5]}{[0.1/ 0.5]}$
$= 5.40$

$pH = 14 - 5.40$
$= 8.60$

pH = 8.60